

Movie Recommendation System – Deliverables

This document outlines the deliverables for the Movie Recommendation System project. The system recommends movies based on user preferences using Machine Learning techniques, including collaborative filtering and content-based filtering.

Project Objective:

Develop a movie recommendation system that suggests relevant movies to users based on: User ratings and interactions Movie genres and metadata Preference similarity using Machine Learning

Component	Technology Used
Programming Language	Python
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn
UI Framework	Streamlit
Dataset	MovieLens Dataset

Key Deliverables: **Jupyter Notebook** Data loading and preprocessing Cleaning missing or duplicate entries Feature engineering and similarity calculations Model training and recommendation generation **Recommendation Logic** Collaborative filtering using user-item similarity Content-based filtering using movie genres Hybrid recommendation strategy for improved accuracy **Streamlit User Interface** User-friendly input section for favorite movies or genres Display top 5 recommended movies Interactive and responsive design **Optional Enhancement** Sentiment-based filtering using movie reviews Improve recommendations using audience feedback analysis

System Workflow:

1. Collect and preprocess MovieLens dataset.
2. Build collaborative filtering recommendation model.
3. Integrate content-based filtering using genres.
4. Accept user preferences through Streamlit UI.
5. Generate and display top 5 movie recommendations.

Expected Output:

The final system will provide personalized movie recommendations in real-time, improving user experience through Machine Learning-based prediction techniques.